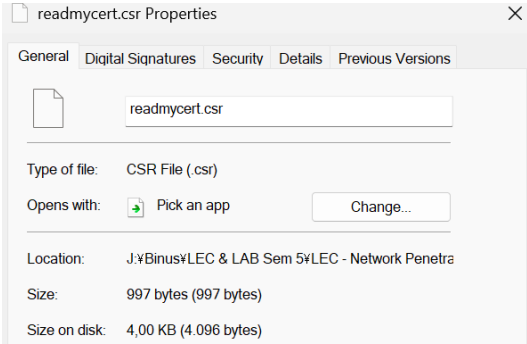


Medium – ReadMyCert

A. The Case

Dalam kasus ini, kita diberi sebuah file "readmycert.csr"



B. Finding the Flag

Biasanya file certificate (.csr) diencode menggunakan Base64.

Menggunakan notepad, kita bisa melihat isi file certificate tersebut:

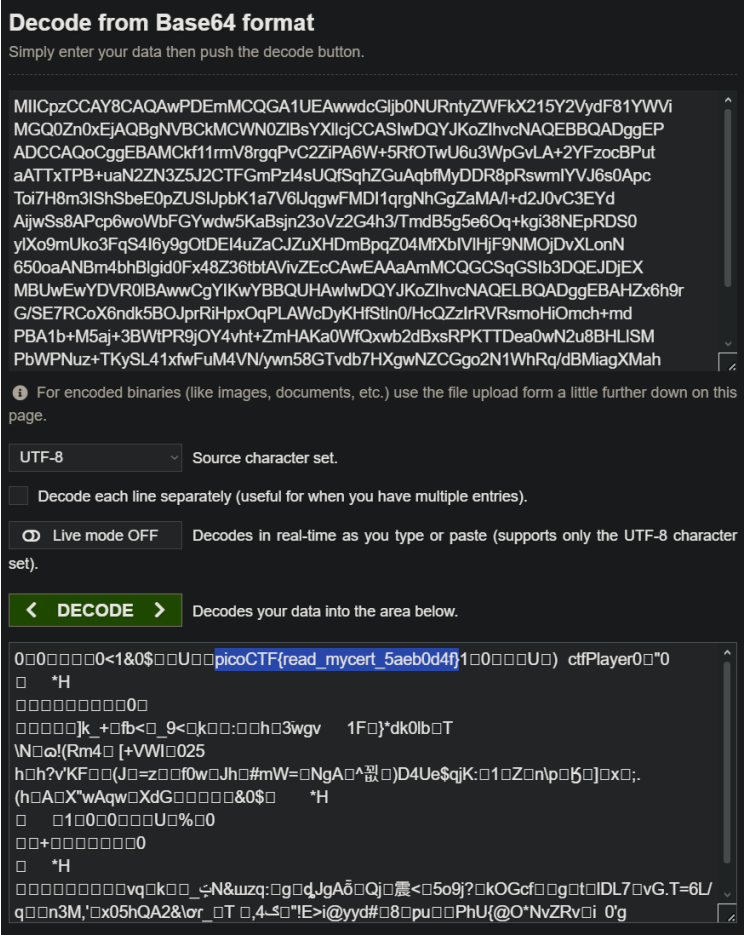
```
-----BEGIN CERTIFICATE REQUEST-----
MIICpzCCAY8CAQAwPDEmMCQGA1UEAwdcG1jb0NURntyZWfkX215Y2VydF81YWVi
MGQ0Zn0xEjAQBgNVBCKMCMW0ZlBsYX11cjCCASIWdQYJKoZIhvcNAQEBBQADggEP
ADCCAQoCggEBAMckf11rmV8rgqPvC2ZiPA6W+5RfOTwU6u3WpGvLA+2YFzocBPut
aATTxTPB+uaN2ZN3Z5J2CTFGmPzI4sUQfSqhZGuAqbFMyDDR8pRswmIYVJ6s0Apc
Toi7H8m3IShSbeE0pZUSIjpbK1a7V6lJqgwFMDI1qrgNhGgZaMA/1+d2J0vC3EYd
AijwSs8APcp6woWbFGYwdw5KaBsJn23oVz2G4h3/TmdB5g5e60q+kgi38NEpRDS0
ylXo9mUko3FqS4I6y9g0tDEI4uZaCJZuXHDmBpqZ04MfXbIVlHjF9NMOjDvXLonN
650oaANBm4bhB1gid0Fx48Z36tbtAVivZEcCAwEAAaAmMCQGCsqGSib3DQEJDjEX
MBUwEwYDVR0lBAwwCgYIKwYBBQUHAIwDQYJKoZIhvcNAQELBQADggEBAHZx6h9r
G/SE7RCox6ndk5B0JprRiHpxOqPLAWcDyKHfStIn0/HcQZzIrRVRsmoHi0mch+md
PBA1b+M5aj+3BwtPR9jOY4vht+ZmHAKa0WfQxwb2dBxsRPKTTDea0wN2u8BHL1SM
PbWPNuz+TKySL41xfwFuM4VN/ywn58GTvdb7HXgwNZCGgo2N1WhRq/dBmiagXMah
yb6gX4erugCu61T5tyD80hgsNBjaqyIdy/whRfC/Pmn3QHmdkqB5ZCPezwb2OLm4
5RDgv3WOB5q0BofoUGHVq757QE8qhL3oTvV2WlLoi3YWaZKJMCeR3vnH92cKC10v
FxdQuLOH8GMv17U=
-----END CERTIFICATE REQUEST-----
```

Ada dua cara untuk melihat isi file certificate ini:

- Cara 1: menggunakan openssl command
 - Command: openssl req -in readmycert.csr -noout -text

```
J:\Binus\LEC & LAB Sem 5\LEC - Network Penetration Testing\Assignment#4>openssl req -in readmycert.csr -noout -text
Certificate Request:
Data:
  Version: 1 (0x0)
  Subject: CN=picoCTF{read_mycert_5aeb0d4f}, name=ctfPlayer
  Subject Public Key Info:
    Public Key Algorithm: rsaEncryption
    Public-Key: (2048 bit)
    Modulus:
      00:c0:a4:7f:5d:6b:99:5f:2b:82:a3:ef:0b:66:62:
      3c:0e:96:fb:94:5f:39:3c:14:ea:ed:d6:a4:6b:cb:
      03:ed:98:17:3a:1c:04:fb:ad:68:04:d3:c5:33:c1:
      fa:e6:8d:d9:93:77:67:92:76:09:31:46:98:fc:c8:
      e2:c5:10:7d:2a:a1:64:6b:80:a9:b7:cc:c8:30:d1:
      f2:94:6c:c2:62:18:54:9e:ac:d0:0a:5c:4e:88:bb:
      1f:c9:b7:21:28:52:6d:e1:34:a5:95:12:20:9a:5b:
      2b:56:bb:57:a9:49:aa:0c:05:30:32:35:aa:b8:0d:
      84:68:19:68:c0:3f:97:e7:76:27:4b:c2:dc:46:1d:
      02:28:f0:4a:cf:00:3d:ca:7a:c2:85:9b:14:66:30:
      77:0e:4a:68:1b:23:9f:6d:e8:57:3d:86:e2:1d:ff:
      4e:67:41:e6:0e:5e:e8:ea:be:92:08:b7:f0:d1:29:
      44:34:b4:ca:55:e8:f6:65:24:a3:71:6a:4b:82:3a:
      cb:d8:0e:b4:31:08:e2:e6:5a:08:96:6e:5c:70:e6:
      06:9a:99:d3:83:1f:5d:b2:15:94:78:c5:f4:d3:0e:
      8c:3b:d7:2e:89:cd:eb:9d:28:68:03:41:9b:86:e1:
      06:58:22:77:41:71:e3:c6:77:ea:d6:ed:01:58:af:
      64:57
    Exponent: 65537 (0x10001)
  Attributes:
    Requested Extensions:
      X509v3 Extended Key Usage:
        TLS Web Client Authentication
  Signature Algorithm: sha256WithRSAEncryption
  Signature Value:
    76:71:ea:1f:6b:1b:f4:84:ed:10:a8:5f:a9:dd:93:90:4e:26:
    9a:d1:88:7a:71:3a:a3:cb:01:67:03:c8:a1:df:4a:d9:67:d3:
    f1:dc:41:9c:c8:ad:15:51:b2:6a:07:88:e9:9c:87:e9:9d:3c:
    10:35:6f:e3:39:6a:3f:b7:05:6b:4f:47:d8:ce:63:8b:e1:b7:
    e6:66:1c:02:9a:d1:67:d0:c7:06:f6:74:1c:6c:44:f2:93:4c:
    37:9a:d3:03:76:bb:c0:47:2e:54:8c:3d:b5:8f:36:ec:fe:4c:
    ac:92:2f:8d:71:7f:01:6e:33:85:4d:ff:2c:27:e7:c1:93:bd:
    d6:fb:1d:78:30:35:90:86:82:8d:8d:d5:68:51:ab:f7:41:32:
    26:a0:5c:c6:a1:c9:be:a0:5f:87:ab:ba:00:ae:eb:54:f9:b7:
    20:fc:d2:18:2c:34:18:da:ab:22:1d:cb:fc:21:45:f0:bf:3e:
    69:f7:40:79:9d:92:a0:79:64:23:de:cf:06:f6:38:b9:b8:e5:
    10:c6:bf:75:8e:07:9a:b4:06:87:e8:50:68:55:ab:be:7b:40:
    4f:2a:84:bd:e8:4e:f5:76:5a:52:e8:8b:76:16:69:99:09:30:
    27:91:de:f9:c7:f7:67:0a:0b:53:af:17:17:50:b8:b3:87:f0:
    63:2f:97:b5
```

➤ Cara 2: menggunakan Base64 decoder



- Flag yang didapatkan: picoCTF{read_mycert_5aeb0d4f}

Setelah kita masukkan flag tersebut, ternyata flag yang kita dapatkan sudah benar.

